Vignesh Jagathese

Visiting Assistant Professor, NDSU

(732) 447 3062
vignesh.jagathese@ndsu.edu
vigj.net
in *vignesh-jagathese*
I am a US Citizen.

Academic Appointments

2026-Present **Visiting Assistant Professor**, *North Dakota State University (NDSU)*
Postdoctoral Mentor: Janet Page

Education

May 2026 **PhD, Mathematics**, *University of Illinois Chicago (UIC)*
Permanence Properties of Quasi-F-Singularities over Arbitrary Fields, Advised By Kevin Tucker

April 2020 **Masters of Science, Mathematics**, *University of Michigan (UofM)*

April 2020 **Bachelor of Science, Honors Mathematics**, *University of Michigan (UofM)*

Mathematical Interests

Commutative Algebra, Algebraic Geometry, F-Singularities, Quasi-F-Splittings, F-Pure and Plus-Pure Thresholds, Singularities via Cartier Operators

Work Experience

2020-2021 **VECO Power Trading**, *Quantitative Trader and Researcher*
Developed models to predict day ahead energy prices, written using R, PHP, MySQL and Python.

Summer 2019 **Chicago Trading Company**, *Quantitative Trading Intern*

2017-2018 **University of Michigan Solar Car Team**, *Operations Director*

Publications and Preprints

[3] **Bounds on the plus-pure thresholds of some hypersurfaces in (ramified) regular rings**, *(2026), with Marta Benozzo, Vaibhav Pandey, Pedro Ramirez Moreno, Karl Schwede, and Prashanth Sridhar*, 2509.07217, (**Mathematische Zeitschrift** 313 (2). Article no. 32)

[2] **On quasi-$\mathbf{F}$-purity of excellent rings**, *(2026)*, 2503.10778, (***Communications in Algebra*** 54 (4). 1760-1771)

[1] **On $\mathbf{F}$-pure thresholds and quasi-$\mathbf{F}$-purity of hypersurfaces**, *(2025), with Jack J Garzella*, 2509.06211, Submitted

Awards and Grants

January 2027 **Collaborate@ICERM**, Funded stay at ICERM as member of a collaboration investigating plus-pure thresholds.

2025 **UIC Award for Graduate Research**, The Graduate College's longest-running support for research by graduate students at UIC. Comes with cash stipend

2023 **UIC Professor Herbert Alexander Award**, Departmenal award, with cash stipend

2022-2026 **NSF RTG Fellowship**, Research and Training (pre-doctoral) Grant

Spring 2019 **NSF REU**, Research Experience for Undergraduates administered through UofM

Teaching

Instructor of Record Positions

Fall 2026 **MATH 144**, *Mathematics For Business*, NDSU

Summer 2026 **SEMW 108**, *Summer Enrichment Math Workshop–Intermediate Algebra*, UIC

Spring 2025 **MATH 109**, *College Algebra Workshop*, UIC

Fall 2023 **MATH 108**, *Intermediate Algebra*, UIC

Spring 2023 **MATH 104**, *Mathematical Reasoning Workshop*, UIC

Fall 2022 **MATH 109**, *College Algebra Workshop*, UIC

Administrative Positions

Fall 2025 **Teaching Assistant Coordinator**, UIC
I worked with the Associate Head of Instruction to train and evaluate new TAs in the department.

Teaching Assistant Positions

Spring 2025 **MATH 110**, *College Algebra*, UIC

Spring 2023 **MATH 105**, *Mathematical Reasoning*, UIC

Fall 2022 **MATH 110**, *College Algebra*, UIC

Fall 2021 **MATH 110**, *College Algebra*, UIC

Fall 2021 **MATH 125**, *Elementary Linear Algebra for Business*, UIC

Grading Positions

Fall 2023 **MATH 516**, *A Second Course in Abstract Algebra I*, UIC

Winter 2020 **MATH 417**, *Matrix Algebra*, UofM

Winter 2019 **MATH 452**, *Multivariable Analysis*, UofM

Invited Talks

10/2026 **NDSU Algebra and Discrete Math Seminar (multi-lecture series)**

07/2026 **UIC Young Scholars Program**

04/2026 **AMS Sectional at NDSU: Topics in Commutative Algebra**

02/2026 **University of Utah Commutative Algebra Seminar**

01/2026 **JMM special session: Advances in Commutative Algebra**

11/2025 **University of Missouri Algebra Seminar**

10/2025 **University of Nebraska Commutative Algebra Seminar**

09/2025 **Purdue Commutative Algebra Seminar**

05/2025 **UC San Diego Algebraic Geometry Seminar**

03/2024 **UIC Incoming Students Colloquium** ***(Also 03/2025)***

01/2023 **UIC Graduate Student Colloquium** ***(Also 04/2024, 01/2025)***

4/2022 **UIC Undergraduate Math Club** ***(Also 08/2023)***

Service

Journal Referee
Refereed for Mathematische Zeitschrift.

2026 **NDSU $\mathbf{F}$-Singularities Reading Group Co-Organizer**
Organized a reading seminar on F-singularities for graduate students, alongside Janet Page.

2024-2025 **UIC Directed Reading Program Advisor**
- **(Spring 2025)** Advised statistics undergraduate Juan Garcia on introductory abstract algebra.
- **(Fall 2024)** Advised electrical engineering and physics undergraduate Caleb Williams on applications of category theory.
- **(Spring 2024)** Advised mathematics undergraduate Max Nguyen on sheaf cohomology, specifically on computations via the Čech complex.

2023-2024 **UIC Commutative and Homological Algebra Graduate Seminar Organizer**
Founded and organized a graduate seminar on active research topics in algebra.

2022-2023 **UIC Mathematics Graduate Student Association Secretary**

Outreach

2022-2026 **Math Circles of Chicago Teacher**
Taught weekend sessions and was an instructor for their summer programs in 2023 and 2026.

11/2024 **Sonia Kowalevsky Math Day Organizer**
Helped organize an annual outreach program to help spark young womens' interest in math.

Summer 2024 **UIC Young Scholars Program Instructor**
Taught and helped organize a free summer program for Chicago high school students at UIC.

2018-2020 **Wolverine Pathways Instructor**

4/2019 **MMaths Math Competition Organizer**